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```

```
-- SQL DIVISION OPERATOR REAL CASE
```

```
-- Use Case: Employees who completed ALL required courses
```

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```

```
DROP DATABASE IF EXISTS division_lab;
```

```
CREATE DATABASE division_lab;
```

```
USE division_lab;
```

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```
-- 1. CREATE TABLES
```

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```

```
CREATE TABLE employees (  
    employee_id INT AUTO_INCREMENT PRIMARY KEY,  
    employee_name VARCHAR(100) NOT NULL,  
    department VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE required_courses (  
    course_id INT AUTO_INCREMENT PRIMARY KEY,  
    course_name VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE employee_courses (  
    employee_id INT NOT NULL,
```

```
course_id INT NOT NULL,  
completion_date DATE NOT NULL,  
PRIMARY KEY (employee_id, course_id),  
FOREIGN KEY (employee_id) REFERENCES employees(employee_id),  
FOREIGN KEY (course_id) REFERENCES required_courses(course_id)  
);
```

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```

```
-- 2. POPULATE DATA
```

```
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```

```
INSERT INTO employees (employee_name, department)
```

```
VALUES
```

```
('Aung Aung', 'IT'),
```

```
('Su Su', 'HR'),
```

```
('Kyaw Kyaw', 'Finance'),
```

```
('Mya Mya', 'IT'),
```

```
('Ko Ko', 'Operations');
```

```
INSERT INTO required_courses (course_name)
```

```
VALUES
```

```
('Cyber Security Awareness'),
```

```
('Data Privacy'),
```

```
('Workplace Safety');
```

-- Aung Aung completed ALL required courses

INSERT INTO employee_courses VALUES

(1, 1, '2026-01-10'),

(1, 2, '2026-01-11'),

(1, 3, '2026-01-12'),

-- Su Su completed only 2 courses

(2, 1, '2026-01-10'),

(2, 2, '2026-01-11'),

-- Kyaw Kyaw completed ALL required courses

(3, 1, '2026-01-15'),

(3, 2, '2026-01-16'),

(3, 3, '2026-01-17'),

-- Mya Mya completed only 1 course

(4, 1, '2026-01-20'),

-- Ko Ko completed ALL required courses

(5, 1, '2026-01-22'),

(5, 2, '2026-01-23'),

(5, 3, '2026-01-24');

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-- 3. VIEW SAMPLE DATA

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79 • **SELECT * FROM** employees;

Result Grid			
Filter Rows:			
Edit:			
	employee_id	employee_name	department
▶	1	Aung Aung	IT
	2	Su Su	HR
	3	Kyaw Kyaw	Finance
	4	Mya Mya	IT
	5	Ko Ko	Operations
*	NULL	NULL	NULL

80 • **SELECT * FROM** required_courses;

Result Grid	
Filter Rows:	
Edit:	
	course_id
	course_name
▶	1
	2
	3
*	NULL

81 • **SELECT * FROM** employee_courses;

Result Grid			
Filter Rows:			
Edit:			
	employee_id	course_id	completion_date
▶	1	1	2026-01-10
	1	2	2026-01-11
	1	3	2026-01-12
	2	1	2026-01-10
	2	2	2026-01-11
	3	1	2026-01-15
	3	2	2026-01-16
	3	3	2026-01-17
	4	1	2026-01-20
	5	1	2026-01-22
	5	2	2026-01-23
	5	3	2026-01-24
*	NULL	NULL	NULL

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-- 4. DIVISION QUERY

-- Find employees who completed ALL required courses

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```
88 • SELECT
89     e.employee_id,
90     e.employee_name,
91     e.department
92 FROM employees e
93 JOIN employee_courses ec
94     ON e.employee_id = ec.employee_id
95 GROUP BY
96     e.employee_id,
97     e.employee_name,
98     e.department
99 HAVING COUNT(DISTINCT ec.course_id) = (
100     SELECT COUNT(*) FROM required_courses
101 );
```

Result Grid			
Filter Rows:			
Export:  Wrap Cell Content: 			
	employee_id	employee_name	department
▶	1	Aung Aung	IT
	3	Kyaw Kyaw	Finance
	5	Ko Ko	Operations

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-- 5. SHOW COMPLETION COUNT PER EMPLOYEE

-- =====

```

112 • SELECT
113     e.employee_name,
114     COUNT(DISTINCT ec.course_id) AS completed_courses,
115     (SELECT COUNT(*) FROM required_courses) AS required_courses
116 FROM employees e
117 LEFT JOIN employee_courses ec
118     ON e.employee_id = ec.employee_id
119 GROUP BY e.employee_id, e.employee_name;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
employee_name	completed_courses	required_courses	
Aung Aung	3	3	
Su Su	2	3	
Kyaw Kyaw	3	3	
Mya Mya	1	3	
Ko Ko	3	3	

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-- 6. FIND EMPLOYEES WHO HAVE NOT COMPLETED ALL COURSES

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```

125 • SELECT
126     e.employee_id,
127     e.employee_name,
128     e.department
129 FROM employees e
130 LEFT JOIN employee_courses ec
131     ON e.employee_id = ec.employee_id
132 GROUP BY
133     e.employee_id,
134     e.employee_name,
135     e.department
136 HAVING COUNT(DISTINCT ec.course_id) < (
137     SELECT COUNT(*) FROM required_courses
138 );

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
employee_id	employee_name	department	
2	Su Su	HR	
4	Mya Mya	IT	